

NAOMAOHU, China

WMO: 521120

Lat: 43.7537N

Lon: 94.9748E

Elev: 1575

StdP: 13.88

Time Zone: 8.00 (E08)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.7	-2.3	-20.9	1.8	6.2	-17.0	2.3	8.8	20.7	19.6	17.8	17.8	3.5	250	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	103.1	65.5	100.5	64.5	98.1	63.7	67.6	94.7	66.5	93.1	65.4	91.9	7.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	82.6	74.9	57.0	73.3	77.9	54.0	65.7	80.5	33.0	94.9	32.0	93.1	31.1	91.7	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	29.2	24.9	21.1	DB	-8.7	108.7	5.9	2.0	-13.0	110.1	-16.4	111.3	-19.8	112.4	-24.0	113.9
(5)				WB	-9.6	69.3	5.5	1.6	-13.6	70.5	-16.8	71.4	-19.9	72.3	-23.9	73.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	12.9	24.6	43.8	61.2	71.5	82.9	86.8	83.0	71.1	53.8	34.6	16.0	(6)
(7)		DBStd	26.78	7.86	8.58	9.26	9.12	6.89	5.91	5.05	5.35	7.06	7.82	9.07	8.58	(7)
(8)		HDD50	3694	1151	712	238	21	0	0	0	0	51	466	1055		(8)
(9)		HDD65	6415	1616	1132	657	173	23	1	0	26	354	913	1520		(9)
(10)		CDD50	5024	0	0	46	356	665	987	1141	1024	634	168	3	0	(10)
(11)		CDD65	2269	0	0	0	58	224	537	676	559	209	6	0	0	(11)
(12)		CDH74	29866	0	0	2	655	2572	7290	9891	7305	2085	66	0	0	(12)
(13)		CDH80	16365	0	0	0	177	1094	4150	6040	4081	816	7	0	0	(13)
(14)	Wind	WSAvg	8.2	5.1	6.8	8.7	11.1	11.2	10.4	9.4	8.7	8.2	7.2	6.3	5.2	(14)
(15)	Precipitation	PrecAvg	3.3	0.0	0.0	0.2	0.2	0.3	0.6	0.8	0.5	0.3	0.2	0.1	0.0	(15)
(16)		PrecMax	5.3	0.1	0.2	1.5	0.4	1.9	1.5	2.0	1.5	0.7	0.6	0.3	0.3	(16)
(17)		PrecMin	2.2	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	0.9	0.0	0.0	0.4	0.1	0.4	0.4	0.3	0.2	0.2	0.1	0.1	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.8	50.5	72.7	89.9	98.3	105.1	107.7	104.7	95.3	79.8	61.3	39.4	(19)
(20)		0.4%	MCWB	28.9	36.0	48.4	57.3	60.7	65.5	67.9	65.9	60.8	54.1	45.3	30.4	(20)
(21)		2%	DB	31.6	45.1	68.5	84.7	93.3	101.4	104.4	100.8	91.4	75.4	56.1	34.6	(21)
(22)		2%	MCWB	24.9	33.2	46.5	54.7	59.2	64.0	66.6	64.6	60.3	51.9	42.2	27.5	(22)
(23)		5%	DB	27.8	41.6	64.7	81.1	90.1	98.8	101.6	98.1	88.2	71.5	51.5	31.3	(23)
(24)		5%	MCWB	22.2	31.0	44.8	53.6	58.1	63.2	65.5	63.8	58.8	50.3	39.6	25.2	(24)
(25)		10%	DB	24.1	38.0	60.4	77.5	86.6	96.2	98.8	95.3	84.9	67.4	47.5	27.6	(25)
(26)		10%	MCWB	19.2	28.8	42.5	51.7	56.5	62.2	64.6	63.1	57.4	48.3	36.8	22.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.1	36.8	49.4	58.2	61.9	67.9	69.7	68.8	63.1	55.6	46.4	31.2	(27)
(28)		0.4%	MCDWB	36.0	49.7	70.3	86.4	91.6	93.0	98.5	94.5	89.4	76.1	58.7	38.3	(28)
(29)		2%	WB	25.3	33.4	47.1	55.9	60.3	66.1	68.2	66.9	61.3	53.2	42.9	27.8	(29)
(30)		2%	MCDWB	31.3	44.7	66.8	81.6	90.4	92.8	96.1	93.2	86.6	72.5	55.5	34.3	(30)
(31)		5%	WB	22.2	31.2	45.1	54.1	58.8	64.9	67.1	65.5	59.8	51.1	40.0	25.4	(31)
(32)		5%	MCDWB	27.6	41.1	63.3	78.8	87.7	92.2	94.3	91.4	84.7	69.7	51.0	31.1	(32)
(33)		10%	WB	19.3	29.0	42.9	52.3	57.3	63.8	66.0	64.4	58.3	49.0	37.3	22.6	(33)
(34)		10%	MCDWB	23.9	37.6	59.6	76.0	84.0	91.3	92.6	90.1	82.7	66.7	47.0	27.2	(34)
(35)	Mean Daily Temperature Range		MDBR	17.8	20.9	24.2	23.1	22.1	20.7	20.3	21.2	22.9	22.4	18.1	15.9	(35)
(36)		5% DB	MCDBR	19.1	23.6	27.4	27.1	27.4	25.0	23.7	24.6	26.1	26.3	21.7	17.5	(36)
(37)		5% DB	MCWBR	14.1	15.1	14.5	12.6	12.3	9.5	8.8	9.6	10.9	13.0	12.6	12.4	(37)
(38)		5% WB	MCDBR	19.0	23.1	26.0	24.0	23.9	21.2	20.6	20.6	23.7	23.8	21.1	17.2	(38)
(39)		5% WB	MCWBR	14.0	14.8	13.8	11.7	10.9	8.0	7.2	8.0	10.0	12.1	12.4	12.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.316	0.353	0.393	0.437	0.442	0.461	0.456	0.426	0.392	0.366	0.337	0.315	(40)	
(41)		taud	2.221	2.159	2.065	2.009	2.059	2.095	2.151	2.217	2.244	2.231	2.241	2.223	(41)	
(42)		Ebn at Noon	250	259	264	262	264	259	259	263	262	252	239	237	(42)	
(43)		Edh at Noon	29	36	46	53	51	50	47	42	38	34	28	26	(43)	
(44)	All-Sky Solar Radiation	RadAvg	701	1040	1523	1924	2245	2230	2175	1992	1687	1228	791	597	(44)	
(45)		RadStd	37	51	68	98	103	134	89	83	61	36	48	49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.83	N/A	N/A	+1.17	-1.84	-2.63	N/A	N/A	+263	+182

Nomenclature: See separate page